INCH-POUND

A-A-53884A
30 September 1992
SUPERSEDING
A-A-53884
19 June 1989

COMMERCIAL ITEM DESCRIPTION

GOLD ALLOY, CASTING, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers dental casting alloys based on gold and used in the fabrication of dental restorations and appliances.

Salient characteristics:

The gold casting alloy shall be in accordance with the American National Standards Institute/American Dental Association Specification (ADA) No. 5 for dental casting gold alloy with the following exceptions and additional requirements.

Types: The dental casting alloy shall be of the following types:

Type I - Soft
Type III - Hard
Type IV - Extra Hard

The gold casting alloy shall be furnished in the above type as specified in the procurement document.

Impurities. The gold casting alloy shall contain not more than 500 parts per million total (ppm) of aluminum, boron, iron, magnesium, manganese, nickel, lead, silicon and tin.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

Ingot. The dental casting alloy shall be furnished in ingot form. Each individual ingot shall weigh 2 pennyweight (dwt) with a tolerance of minus 1.0 percent. Ten ingots shall have a total weight of 20 dwt with a tolerance of minus 0.1 percent. As an alternate, each ingot shall weigh 1 dwt with a tolerance of minus 1.0 percent. Twenty 1 dwt ingots shall have a total weight of 20 dwt with a tolerance of minus 0.1 percent.

Physical properties. In addition to the requirements of ADA Specification No. 5 the gold alloy shall meet the following requirements:

Type	Gold and Metals of Platinum Group (minimum %)	Vickers Number		
		Quenched Min	Max	Hardened Min
		HVI	HVI	HVI
I	83	50	90	---
III	78	120	150	---
IV	75	150	---	220

Melting Range

	Solidus	Liquids
All Types	Min °C	Max °C
	800	1050

Color. The color of the casting alloy shall be gold.

Finish. The ingots shall have a bright finish free of tarnish or other discolorations.

Identification marking. Each ingot shall be stamped with the name or registered trademark of the manufacturer or contractor.

Certification. The supplier shall furnish proof at time of award that the dental casting gold alloy is the same in quality and brand as that appearing on the American Dental Association current list of certified dental casting alloys. Compliance with the certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

Workmanship. The gold shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One ingot weighing 2 pennyweight, or two ingots each weighing 1 pennyweight, of the type specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspection lot.

Initial inspection lot. The initial inspection lot shall be the first 10,000 units or less of one or more melts of production of the same type of alloy for a delivery.

Balance of inspection lot. The balance of the inspection lot shall consist of every 10,000 units or less of production of the same type of alloy for the balance of the delivery.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:26Z
Check the source to verify that this is the current version before use.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit of product for examination purposes shall be one unit of issue. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 2.5.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests.

Sampling of initial inspection lot. Each initial inspection lot shall be sampled for the determination of chemical composition, physical and mechanical properties. One or more ingots from the first melt and physical test specimens cast from random sampling of the ingots comprising the initial inspection lot shall be sampled and tested for compliance with the specification requirements.

Sampling of the balance of the inspection lot. A melt from each of the subsequent inspection lots shall be sampled for the determination of chemical composition. The analysis shall agree with alloy constituents of the initial inspection lot within 0.25 percent. Impurities shall not exceed the value for the initial inspection lot. If the results of this analysis agree with the composition of the initial inspection lot (± 0.25%), no physical test specimens shall be required. If the results of this analysis do not agree with composition of the initial inspection lot (± 0.25%), sampling and testing as specified for the initial inspection lot shall be repeated. If this determination shows the lot to comply with the composition requirements, sampling on the remainder of the inspection lots shall be performed. The new determination of chemical composition shall form the basis for the acceptance of the succeeding lots. If the results show that the lot does not comply with the requirements, the lot shall be rejected.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:26Z
Check the source to verify that this is the currentversion before use.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers and their respective Item Identifications are covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6520-00-145-0350	Gold Alloy, Casting, Dental, Fine Grain, Soft, 2 Pennyweight
6520-00-145-0176	Gold Alloy, Casting, Dental, Fine Grain, Hard, 2 Pennyweight
6520-00-145-0349	Gold Alloy, Casting, Dental, Fine Grain, Extra Hard, 2 Pennyweight

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6520-2532
Location: c:\word\doc\levin\goldcid

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

   The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE. This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-53884A	30 September 1992

3. DOCUMENT TITLE
GOLD ALLOY, CASTING, DENTAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, [illegible])	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center ATTN: DPSC-MSE	(215) 7372870	444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:26Z
Check the source to verify that this is the current version before use.